Completed Triangle Heuristics and Starting Ellipse Study

22 October 2026

Things to note:

- A summary of the implementations up to commit 9adabf6
- Triangle runs, saved folder and overview (include the config perhaps)
- Preliminary ellipse runs x 3 and the results so far:
  - Same set up as triangle, except now ellipse. Compare these to previous runs: finding the local optima?
  - Imposed spacing limits (are these obeyed? Think about WP7)
  - The M1 test for ellipses (however, will assume all other heuristics are the same). Especially point out seeds 101 and 102. Full run of interest? Suggests that circular is best.
- Possible modifications:
  - WP7 if spacing heuristics are not obeyed
  - Option to constrain to be circular
  - Option for max hole size?
  - Turn smin and bx (left/right) into mm rather than mesh units
  - FUTURE PROOF: synthetic buckling provision
  - Also need to implement the M1 findings
  - Option to output the centre co-ordinates etc of all of the holes in the final state (if not already provided)
  - Option to manually perturb results or test new values (e.g. see that the best after BO evaluation 385 is close to being symmetric about the midline)?

What is the solid fraction of the tested beam? I think it is quite low (e.g. approx 0.2?). This should work a bit better with the increased spacing constraints? CHECK THIS. Also, remember that when split into two (symmetric analysis), then how will the volume fraction be affected?

1 Introduction

Extensive numerical studies were conducted on triangular holes. These could establish various heuristic parameters in the config defaults:

- **Step 2B (M1):**

  `observation_noise_variance = 1e-5`

- **Step 3 (M2):**

  `health_predictive_window = 3`

  `health_predictive_z_max = 2.64653631`

  `health_signal_variance_min = None`

  `health_mean_spread_min = None`

  (`health_nu_min: float = 0.01` leave as this default, this is not applicable to prescribed noise)

- **Step 4 (M4):**

  `ng_rst = 10`

  `nl_rst = 5`

- **Step 5 (M5):**

  `acquisition_weight: 0.7`

  `novelty_weight: 0.3`

- **Step 6**

  `warm_start_policy = "health_reset"` produces superior designs than `always_cold_start` when examined for three seeds.

These numerical studies provide sufficient confidence that hyperparameters have been tuned to their most optimum. The reason why this study was conducted on triangular perforations, is because the end result MBB beam is known. However, the end result/most optimum is not known for ellipses.

The heuristics established in the triangle study were taken forward to the ellipse study. However, more tests are being conducted to establish the M1 and M2 parameters (which may be shape dependent). Additionally, a few more modifications were required to get the ellipse code up to a satisfactory state.

As highlighted in `261014_Modifications_To_BO_Code`, there were two issues WP6 and WP7 associated with the ellipse representation:

- **WP6**: the GP and novelty calculations treat equivalent ellipse encodings as different designs
- **WP7**: the directional clearance approximation may introdcue feasibility errors, but this has not yet been demonstrated.

The visually different ellipse optima are also partly legitimate: different geometries can have almost identical compliance. The proposed work should remove representation artefacts, not force all seeds to converge to one geometry.

1.1 WP6 Code Modifications (implemented, commit 5d2dbcd)

- Add an isolated feature helper which contains the raw ellipse decoding and derives GP/diversity features
- Transform GP inputs consistently and use a differentable deterministic BoTorch input transform
- Make diversity calculations invariant
- Preserve reproducability
- Add focussed WP6 tests

See in the config:

- `ellipse_feature_mode = "raw"`: the original superseeded ellipse design. This uses the original ellipse encoding directly and equivalent representations such as swapping the radii and rotating by 90 degrees are (incorrectly) treated as different GP inputs.
- `ellipse_feature_mode = "shape_matrix"`: the new, updated radii/angle swapping-aware GP.

1.2 WP7 Code Modifications (not yet implemented)

This is a clearance-accuracy audit of the directional SAT approximation. This should establish whether the SAT approximation contributes materially to ellipse feasibility errors. It should change geometry handling only if material clearance-classification errors are demonstrated.

- Add an audit script, without changing production clearance
- Construct a converged numerical reference. Compare the production functions `separating_axis_theorem` and `separating_axis_theorem_batch` against a dense angular support-function search, followed by local angular refinement
- Separate sources of error.
- Persist audit results
- Predefine materiality criteria
- Only change clearance if the audit demonstrates a material problem

1.3 Spacing

Before commit 9adabf6:

- `bx` is symmetric: the usable horizontal interval is `[bx, xsize - bx`
- `smin` is already measured in physical mesh units. Thus `smin=2` requests approximately two element widths of clearance. Values above 1 are accepted
- Burn-in uses strict sampling. It must collect the requested quota from both area strata, or it raises an error.
- The early feasibility check considers only usable area, and does not account for smin, shape compactness, or packing difficulty

With the code in this state, increasing margins or smin can cause very slow burn-in, followed by the generic `"could not generate enough..."` error.

1.3.1 New mathematical feasibility limits

Let:

$$\begin{aligned} W &= x_{size} - bx_{left} - bx_{right} \\ H &= y_{size} - 2by \\ V_{min} &= \max\left((1 - solid_{max})x_{size}y_{size}, nholes \cdot a_{min}\right) \end{aligned} \tag{1}$$

Necessary conditions are:

$$bx_{left} \geq 0, bx_{right} \geq 0, W > 0, H > 0, V_{min} \leq WH \tag{2}$$

A stronger necessary packing check follows by expanding every convex hole by $s_{min}/2$. Feasibility requires:

$$V_{min} + s_{min}\left(\sqrt{\pi}Q - W - H\right) + \frac{(n-1)\pi}{4}s_{min}^2 \leq WH \tag{3}$$

This can be solved numerically for an upper screening bound on s_{min}. Passing it does not prove feasibility, failing it proves that the requested packing cannot exist. There is therefore no useful limit on s_{min} independent of margins, hole count, areas, quality limits and domain size.

The following changes were made:

- Split bx into bx_{left} and bx_{right} fields for OptimizationConfig.
- Make burn-in collection robust. Add a preflight feasibility function to check non-negative margins and inter-hole spacing etc. This avoids randomly generating targets below `nholes x amin` and rejecting them

2 Final Config

This is the config, at the commit 9adabf6

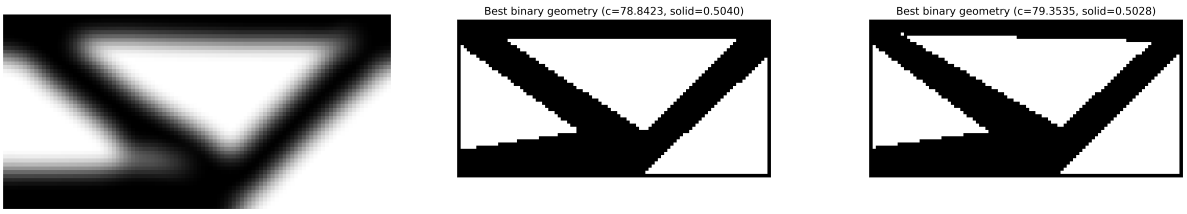
```python
@dataclass
class OptimizationConfig:
    xsize: int = 100
    ysize: int = 50
    bx: float = 1.0
    by: float = 1.0
    amin: float = 50.0
    smin: float = 1.0
    solid_max: float = 0.5
    nholes: int = 3
    htype: int = 0
    ellipse_feature_mode: str = SHAPE_MATRIX_ELLIPSE_FEATURES
    tri_q_min: float = 0.10
    raw_pool_multiplier: int = 3
    ng_rst: int = 10
    nl_rst: int = 5
    max_restart_batches: int = 6
    minimum_restart_distance: float = 0.02
    local_radius: float = 0.05
    local_start_strategy: str = "archive"
    acquisition_weight: float = 0.7
    novelty_weight: float = 0.3
    global_restart_selection_strategy: str = "greedy_dynamic_novelty"
    raster_tolerance: float = 0.005
    emin: float = 1e-9
    e0: float = 1.0
    penal: float = 3.0
    observation_noise_variance: float = 1e-05
    warm_start_policy: str = "health_reset"
    health_nu_min: float = 0.01
    health_signal_variance_min: float | None = None
    health_mean_spread_min: float | None = None
    health_posterior_std_min: float | None = None
    health_predictive_z_max: float = 2.64653631
    health_predictive_window: int = 3
    burn_in: int | None = None
    iterations: int | None = None
    stagnation_patience: int | None = None
    exploration_fraction: float = 0.75
    adaptive_restart_allocation: bool = True
```

```
    seed: int = 0
    output_dir: str | Path | None = None
    audit_reference: bool = True
    bx_left: float | None = None
    bx_right: float | None = None
```

3 Triangle Results

The results in `C:\bo_studies\261021_Triangles`, using the measured hyperparameters. The best geometry is comfortably the minimum (only seeds 108 and 251 have completed at the time of writing). These have found geometries with:

1. The lowest-ever recorded compliance (see the Excel named `261009_One_Shape_Bo_Runs` inside `Bayesian_Optimisation_P`
2. A result that is very similar to the MBB beam result
3. This gives high confidence that the beam is finding the near-global optimum!

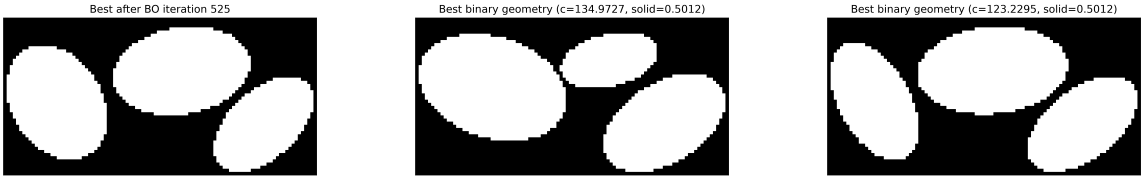


(a) MBB Compliance Classic (b) BO Seed 108 Result (c) BO Seed 251 Result

Figure 1: One very happy Ellie

4 Ellipse Results

These results were run with the same defaults as the triangles (M1 and M2 outputs). This may not be correct. See next section where further analysis conducted. The preliminary results so far are encouraging. These also superseed or match the best ellipse result found previously. At the time of writing, only seeds 108 and 251 have completed.



(a) See the Ellipse folder in 261008 Package Plan folder. Compliance of 122.46 (b) BO Seed 108 Result (c) BO Seed 251 Result

Figure 2: Preliminary Ellipse Results

These clearly belong to the same 'family' as each other. This is interestingly not as simple as drawing an ellipse in the direction of the centreline of the triangle holes above. The fact that these results are similar to each other suggests that the near-global optimum has again been found. To improve these results further, use the properly calibrated M1/M2 values (for ellipses, see next section).

5 Ellipse Hyperparameters

See `C:\bo_studies\Ellipse_Manual_Tests\M1`, and the `261016_BO_Code_Modifications` section 9.2.1 (Step 2A) and section 9.2.2 (Step 2B). These were used to make a decision rule on the observation noise and the health limits. All other heuristic properties are assumed as identical to the triangle study results. Indeed, as the Ellipse results are so similar (see previous section), this is a reasonable conclusion to make.

6 Ellipse with larger spacing

See `C:\bo_studies\261021_Ellipses_Part_2`, NB, this study is much slower than the triangle and ellipse studies with a less restricted dimension space. The behaviour seems far more erratic (possibly multiple local optima, as these are vastly different hole families, see figure below).



(a) Compliance = 476.08 (b) Compliance = 307.20 (c) Compliance = 276.96

Figure 3: Imposed Spacing Ellipse Results

- Should an additional restraint be placed on the min/max perforation size?
- Should the minimum ellipse quality be modified? e.g. see best after 57 evaluations, with a very narrow centre hole
- Must confirm whether these obey the spacing restrictions that were imposed (they do look appropriate).
  - $bx_{left} = 10$
  - $bx_{right} = 5$
  - $by = 5$
  - $s_{min} = 1$

7 Ellipse next

- Are the spacing heuristics obeyed in the previous section? (Does this relate to WP7?)
- Turn xsize, sx, bx and related variables into dimensions, rather than in mesh element metrics.
- Make a modification so that the new ellipse hyperparameters can be accommodated if ellipse is selected.

Proposed runs:

1. Half-beam ellipse, 3 x holes. **5 Tests**
    - Correct dimensions, make use of the updates as listed above
    - Correct loading, based on position of 3pt bending points
    - Correct physics (e.g. E, and other known material properties)
    - Correct the minimum hole volume ratio. Pay attention that modelling half beam!
2. Full-beam ellipse, 5 x holes. **5 Tests**
   See the list above, now for full beam
3. Half-beam ellipse, 3 x holes. Buckling mitigation: synthetic E. **5 tests**
   See list above.
4. Half beam triangle, 3 x holes. As a control or out of interest.
5. With MBB code, run for triangle and determine the baseline compliance.